

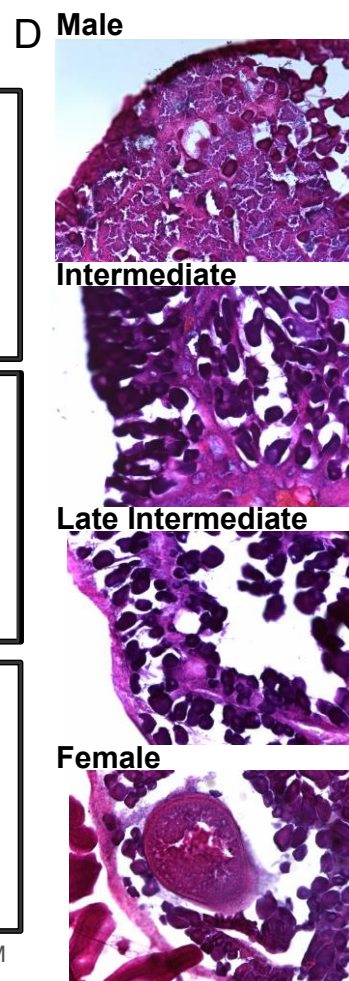
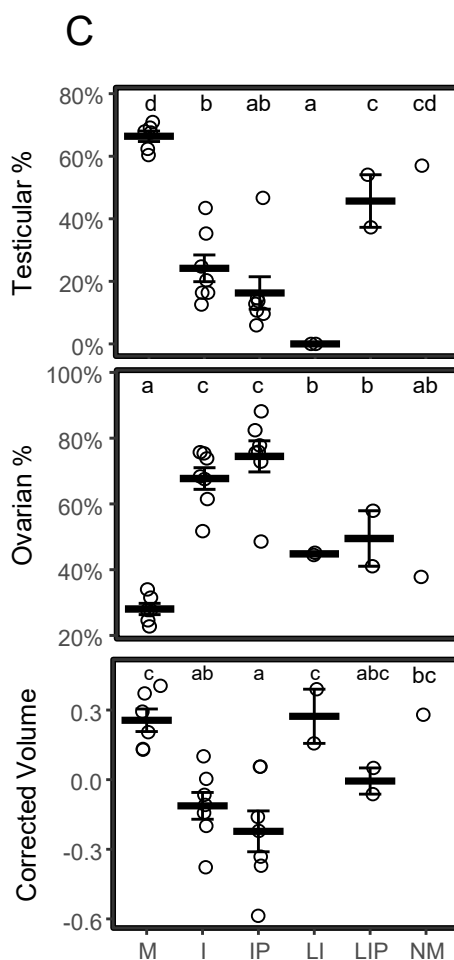
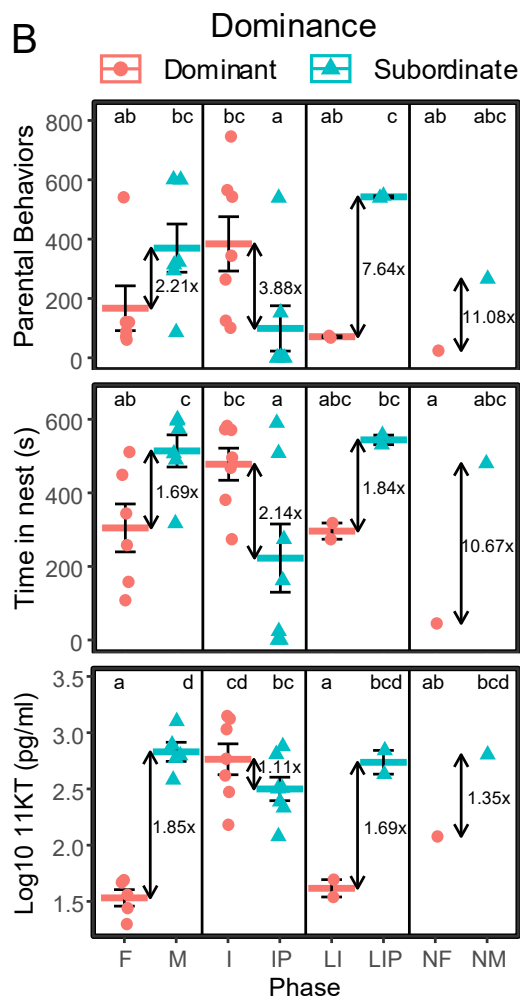
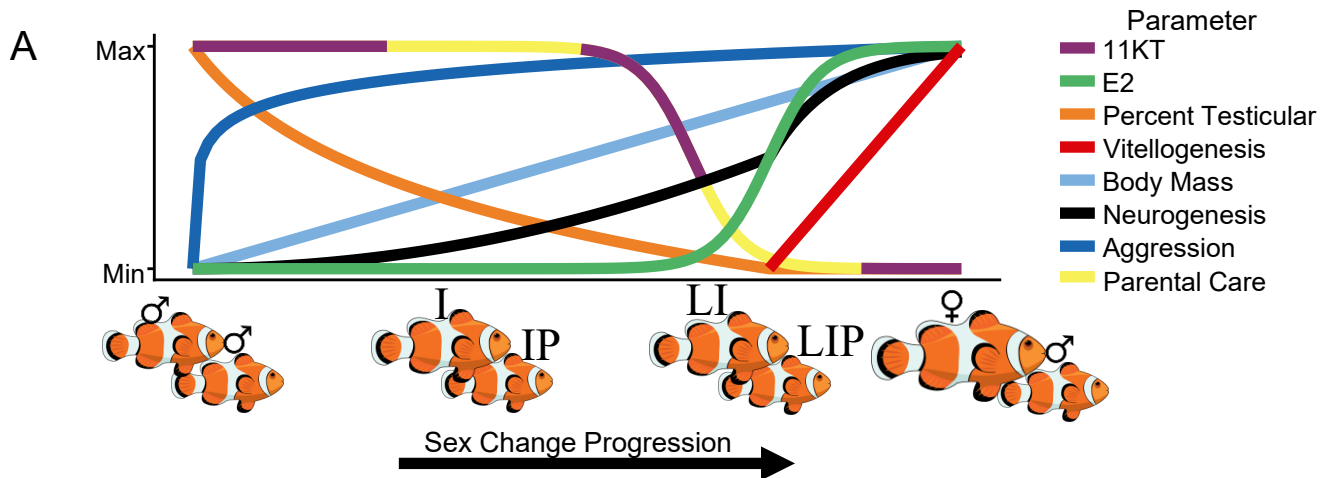


Fig.1. Male, intermediate, and female anemonefish display distinct physiology and behavior. (A) Schematic outlining the behavioral and physiological changes that occur during protandrous sex change in anemonefish. (B) Top panel: Parental behaviors observed during a parental care assay. Middle panel: Time spent in the nest during parental care assay; a proxy for parental care. Bottom Panel: Log<sub>10</sub> transformed serum 11-ketotestosterone. (C) Top panel: Percent of the gonad made of testicular tissue. Middle panel: Percent of the gonad made of ovarian tissue. Bottom panel: Gonadal volume corrected for the individual's length (methods). (D) Example hematoxylin and eosin stains of gonads from the major phases of sex change. In B, plots are subdivided such that the dominant and subordinate individuals from each pair appear together reflected by shape and color.. Arrows indicate fold-difference between the two groups. In B, C, crossbar indicates mean, error bars indicate standard error of the mean (SEM). Letters indicate significance: groups that do not share a letter are significantly different.



Fig.2. Multiomics capture high quality transcriptomes, epigenomes, and spatial transcriptomes, (A) UMAP embeddings of our sn-Multiome data with each cluster numbered. (B) Dotplot showing canonical and data-driven markers for each of our multiome clusters. (C) Heatmap showing the scaled proportion of each neuroanatomical region made of each multiome cluster (see supplement xx for abbreviations). (D) Example projections of multiome clusters onto representative brain sections. The dashed line indicates approximately where we dissected brains for sn-multiomics analysis. Below each section is a sketch of an *A. ocellaris* brain with a line indicating the approximate sectioning plane for the given section.

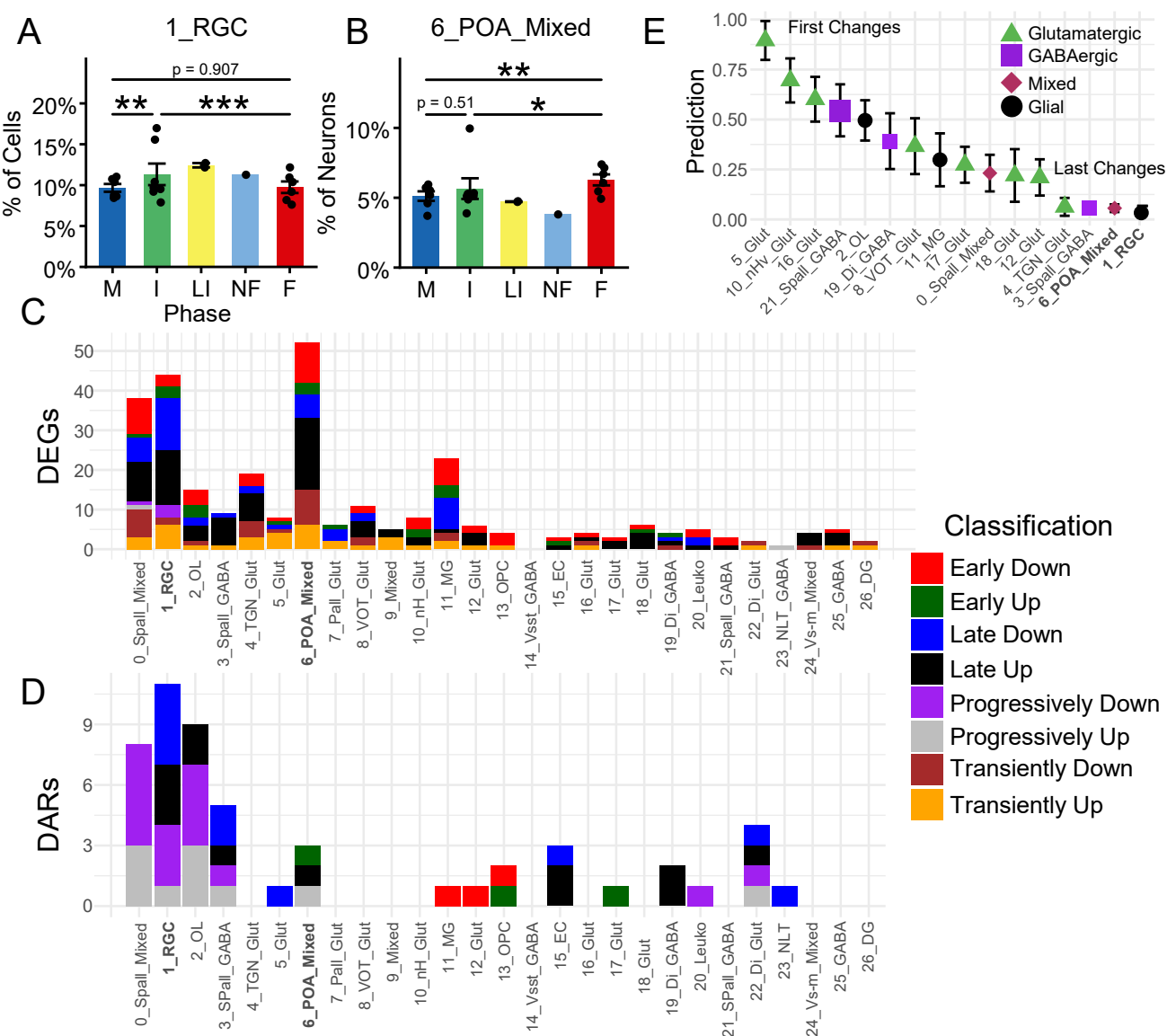


Fig.3. Cells change composition, transcriptome, and epigenome asynchronously during sex change. (A) The percent of all cells in our dataset belonging to 1\_RGC. (B) The percent of all neurons in our dataset belonging to 6\_POA\_Mixed. (C) The number of differentially expressed genes (DEGs) by multiome cluster. (D) The number of differentially accessible regions (DARs) by multiome cluster. (E) Machine learning predictions for the mean sex of intermediates based on their expression of DEGs and DARs by multiome cluster; a proxy for the temporal order of sex change Shape and color represents broad cell type. In C, D, bars are colored based on our classification for the manner in which they change sex (see methods). In A, B, points indicate the proportion for each individual. Asterisks: “\*”:  $p < 0.05$ , “\*\*”:  $p < 0.01$ , “\*\*\*”:  $p < 0.001$ .

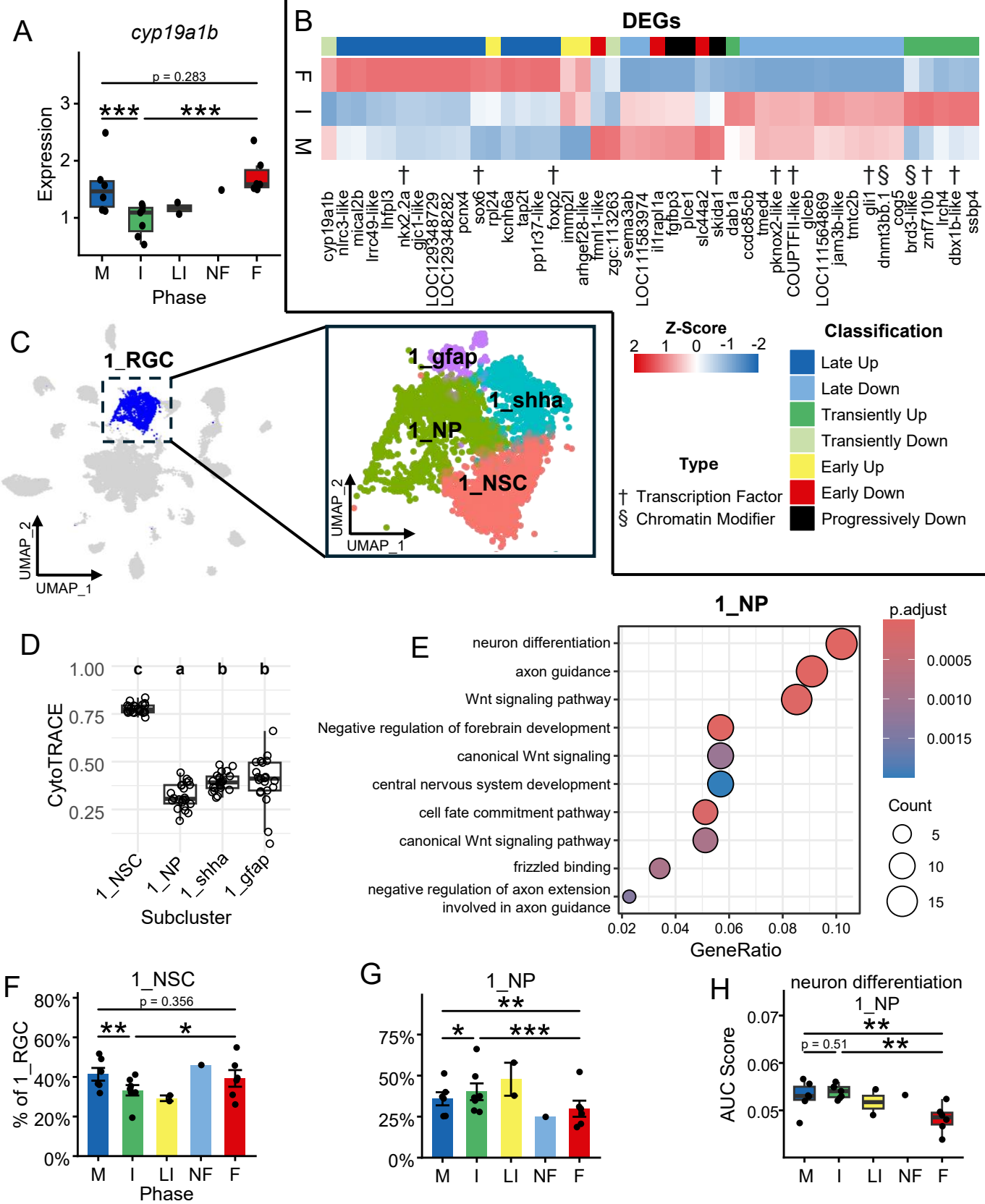


Fig.4. Radial glial cells undergo changes in transcriptome and subtype composition over the course of sex change. (A) Brain aromatase (*cyp19a1b*) expression in 1\_RGC. (B) A heatmap showing all DEGs by phase in 1\_RGC. Cells are colored by z-score of the mean expression for each phase. The top bar indicates the classification for how each gene changes sex, and the symbols below the matrix indicate what kind of gene each DEG is. (C) UMAP showing subclustering of 1\_RGC. (D) Mean CytoTRACE score for each individual by subcluster. A higher score indicates a less mature cell type. Significance is indicated by letters, subclusters who do not share a letter are significantly different from each other. (E) Gene ontology (GO) analysis for the marker genes of 1\_NP. Color indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list. (F) Percent of 1\_RGC composed of 1\_NSC. (G) Percent of 1\_RGC composed of 1\_NP. (H) AUCell score of a gene expression module composed of genes labeled with the GO term “neuron differentiation” in 1\_NP. (I) Schematic describing our model for how neurogenesis changed over the course of sex change. In A, F-H, points indicate the mean for each individual. Asterisks: “\*”:  $p < 0.05$ , “\*\*”:  $p < 0.01$ , “\*\*\*”:  $p < 0.001$ .



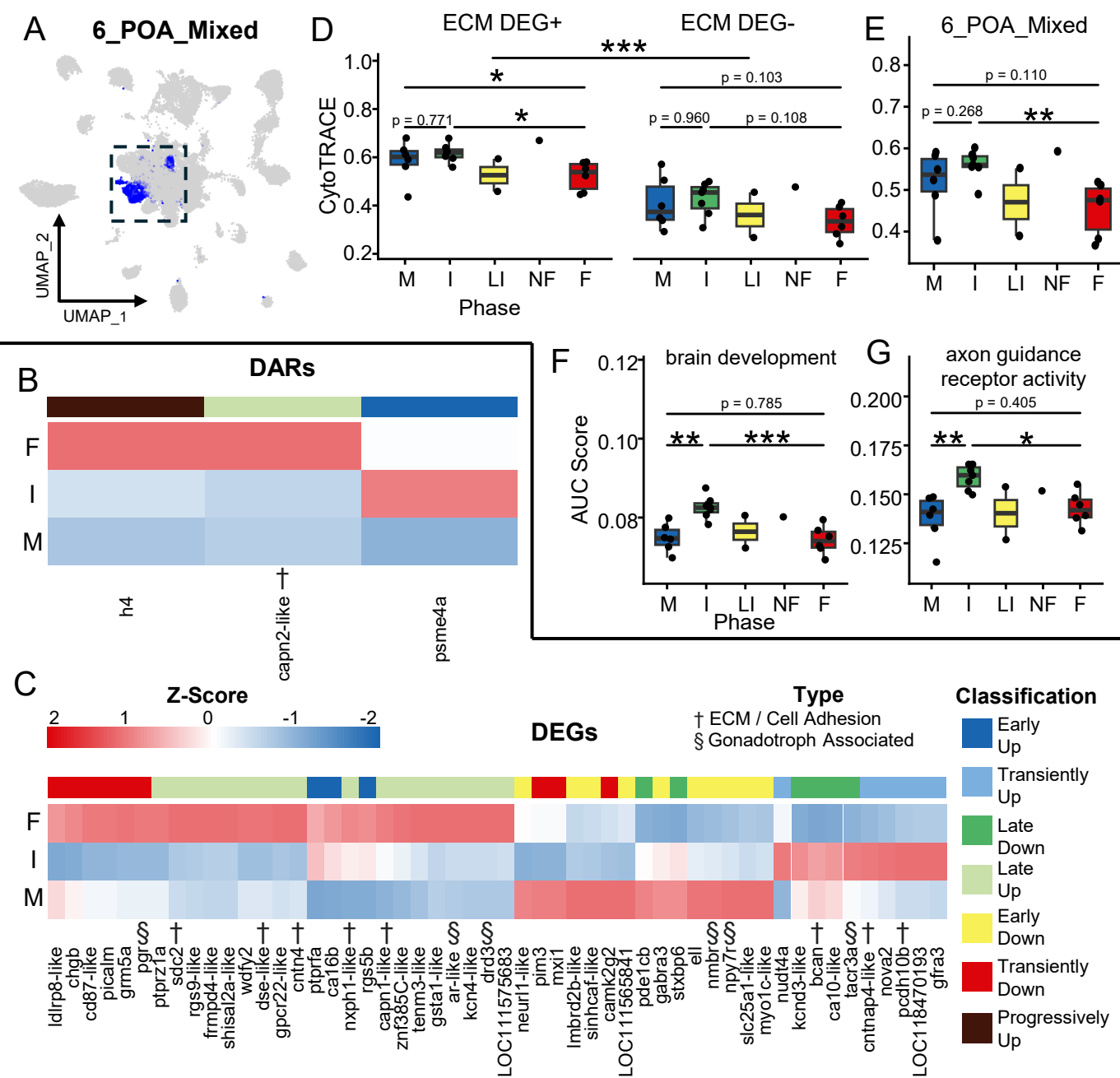


Fig.5. 6\_POA\_Mixed undergoes changes in transcriptome and epigenome consistent with neurogenesis. (A) UMAP projection with 6\_POA\_Mixed highlighted. (B) DARs in 6\_POA\_mixed. (C) DEGs in 6\_POA\_Mixed. In B, C, cells are colored by z-score of the mean accessibility or expression for each phase. The top bar indicates the classification for how each gene or region changes sex, and the symbols below the matrix indicate what kind of gene is differentially accessible or differentially expressed. (D) CytoTRACE score by phase split by whether the cell expresses a DEG associated with the extracellular matrix (ECM-DEG) or not. (E) CytoTRACE score for 6\_POA\_Mixed as a whole. (F) AUCell score for a gene expression module made of genes annotated with the GO term “brain development”. (G) AUCell score for a gene expression module made of genes annotated with the GO term “axon guidance receptor activity”. In D-G, points indicate the mean for each individual. Asterisks: “\*”:  $p < 0.05$ , “\*\*”:  $p < 0.01$ , “\*\*\*”:  $p < 0.001$ .

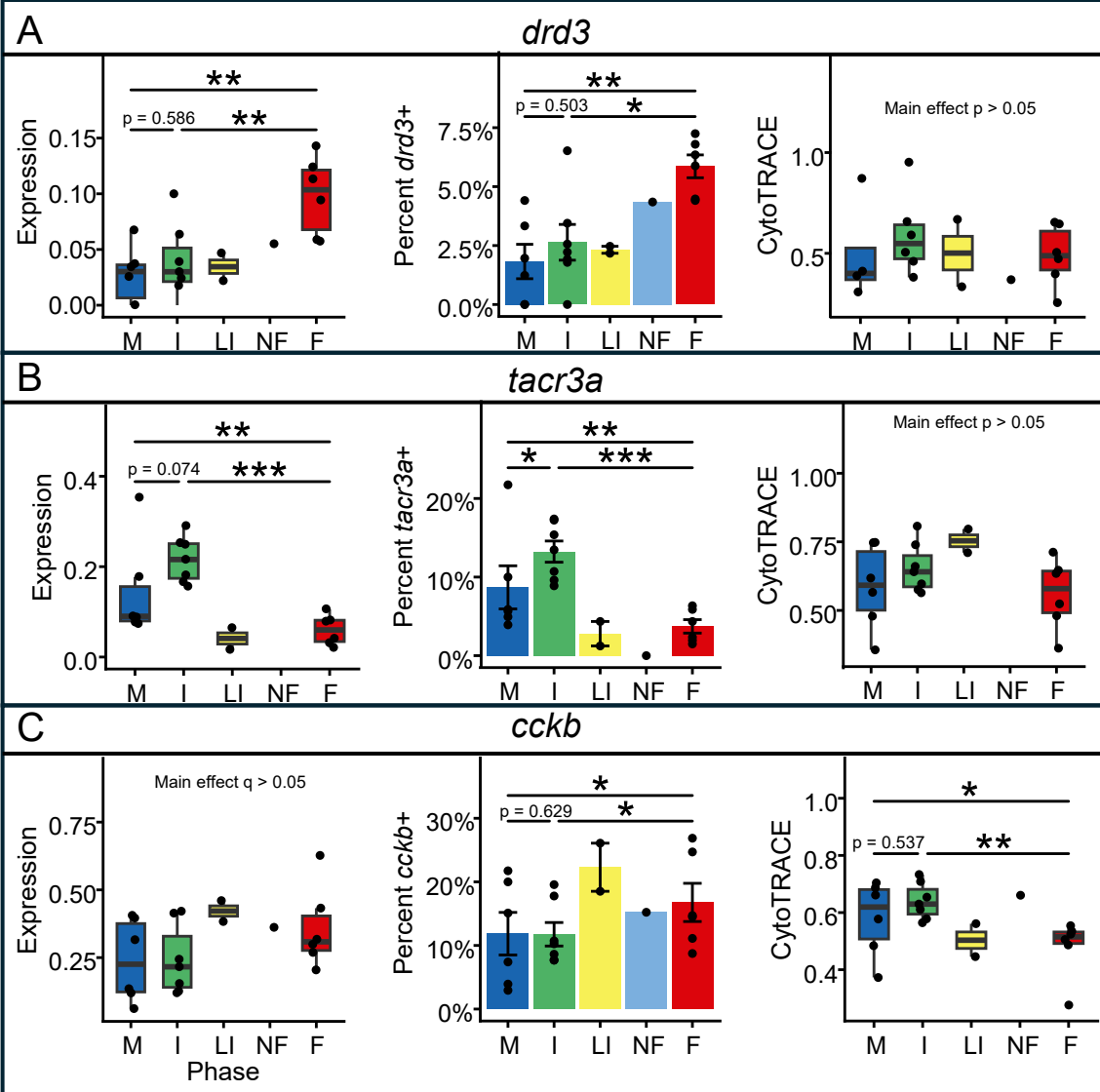
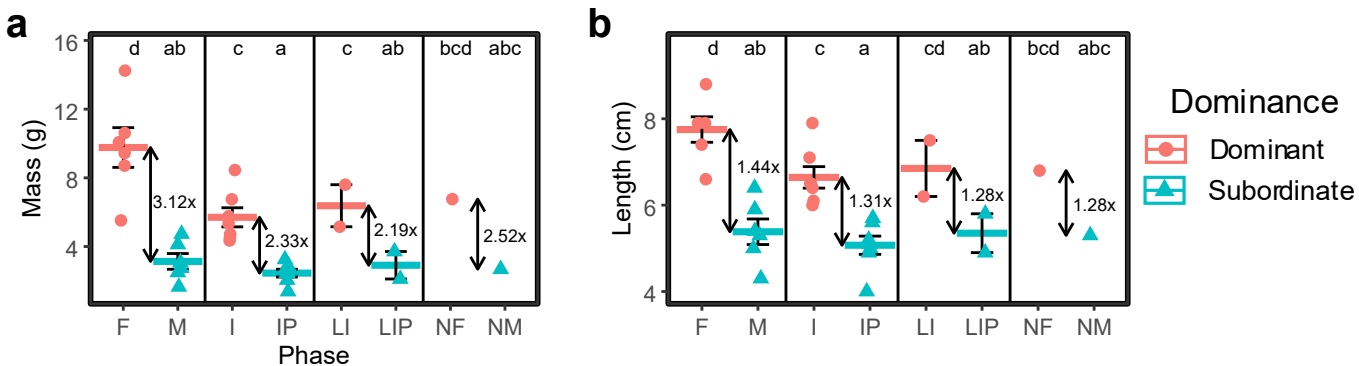
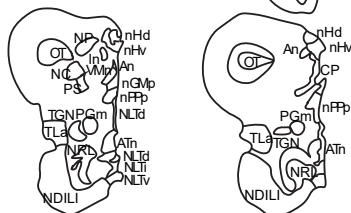


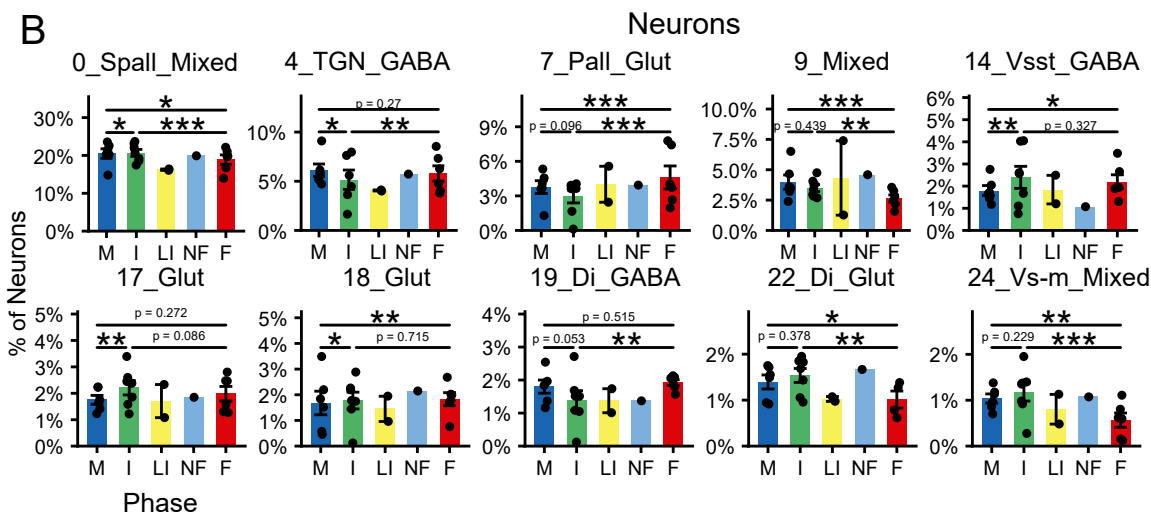
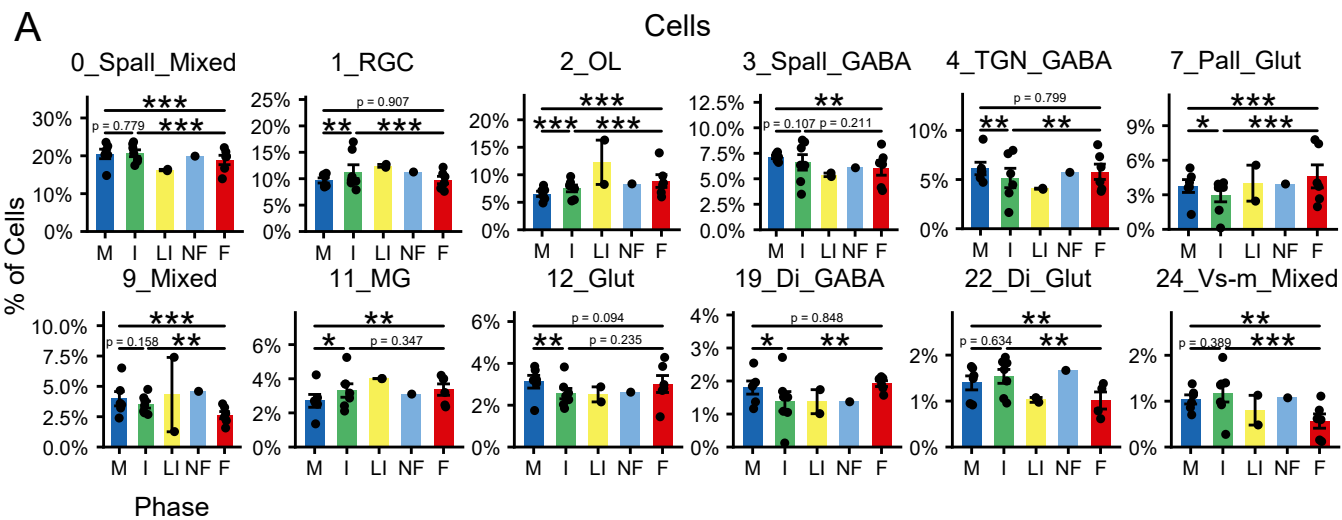
Fig.6. Changes to 6\_POA\_Mixed Drive Differential Regulation of Gonadotrophs. (A) Left panel expression of *drd3* by phase. Middle panel: Percent of 6\_POA\_Mixed cells expressing *drd3* by phase. Right panel: CyoTRACE score for *drd3*+ cells by phase.(B) Left panel expression of *tacr3a* by phase. Middle panel: Percent of 6\_POA\_Mixed cells expressing *tacr3a* by phase. Right panel: CyoTRACE score for *tacr3a*+ cells by phase.(C) Left panel expression of *cckb* by phase. Middle panel: Percent of 6\_POA\_Mixed cells expressing *cckb* by phase. Right panel: CyoTRACE score for *cckb* + cells by phase. Points indicate the mean for each individual. Asterisks: “\*”:  $p < 0.05$ , “\*\*”:  $p < 0.01$ , “\*\*\*”:  $p < 0.001$ . “Main effect Main effect  $p > 0.05$ ” indicates type III tests were not significant so pairwise comparisons were not computed.



Supplementary Fig.1. (A) Body mass by phase. (B) Body length by phase. Plots are subdivided such that the dominant and subordinate individuals from each pair appear together reflected by shape and color.. Arrows indicate fold-difference between the two groups. Crossbar indicates mean, error bars indicate standard error of the mean (SEM). Letters indicate significance: groups that do not share a letter are significantly different..

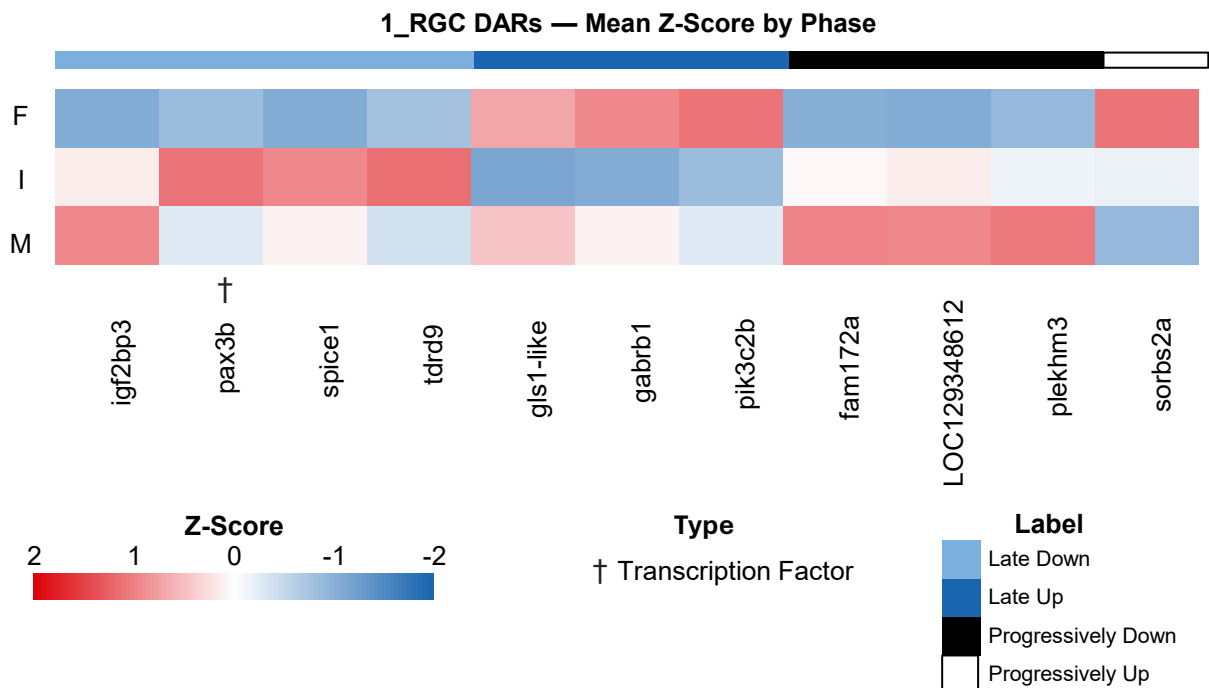


Supplementary Fig.2. Drawings of each section sequenced with VisiumHD with neuroanatomical regions labeled.



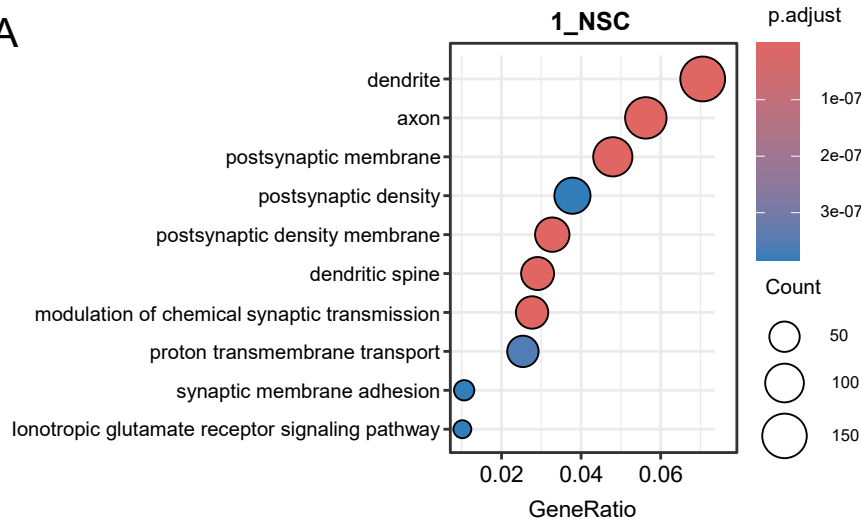
Supplementary Fig.3. (A) Percent of all cells belonging to each multiome cluster. (B) Percent of all neurons belonging to each multiome cluster. Only significant results are shown. Points indicate the mean for each individual. Asterisks: “\*”:  $p < 0.05$ , “\*\*”:  $p < 0.01$ , “\*\*\*”:  $p < 0.001$ .



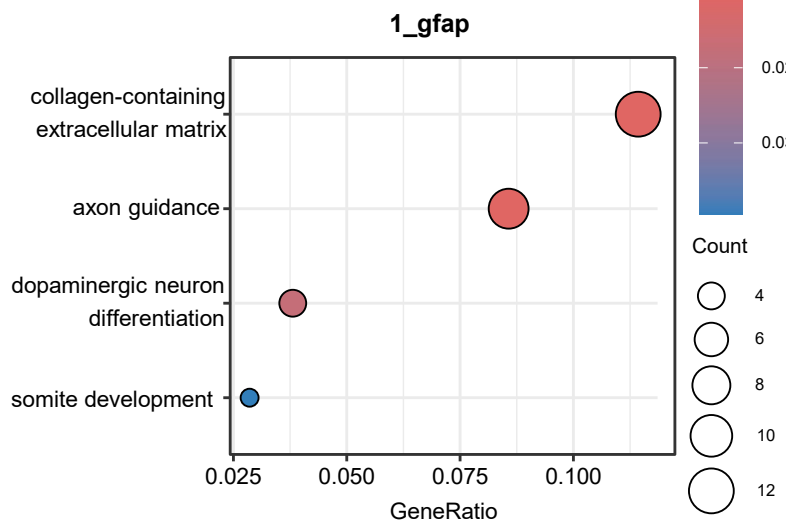


Supplementary Fig.4. DARs in 1\_RGC. Cells are colored by z-score of the mean accessibility or expression for each phase. The top bar indicates the classification for how each gene or region changes sex, and the symbols below the matrix indicate what kind of gene is differentially accessible or differentially expressed. (B)

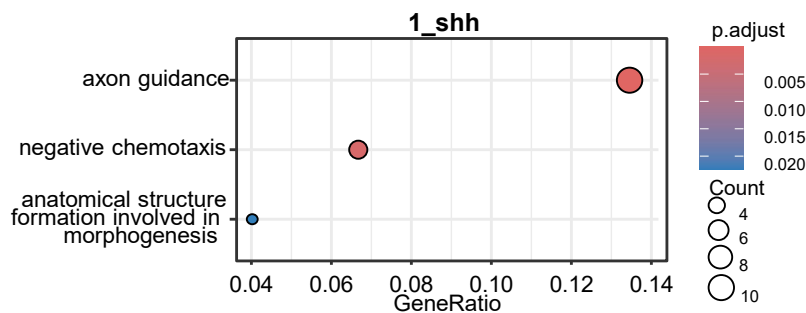
A



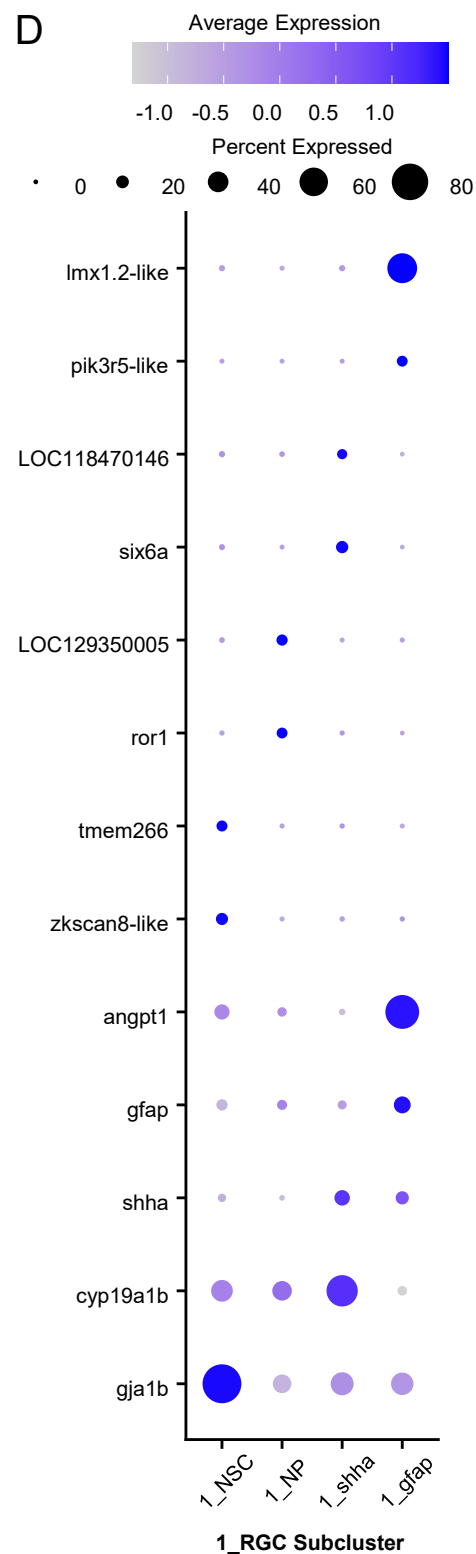
B



C

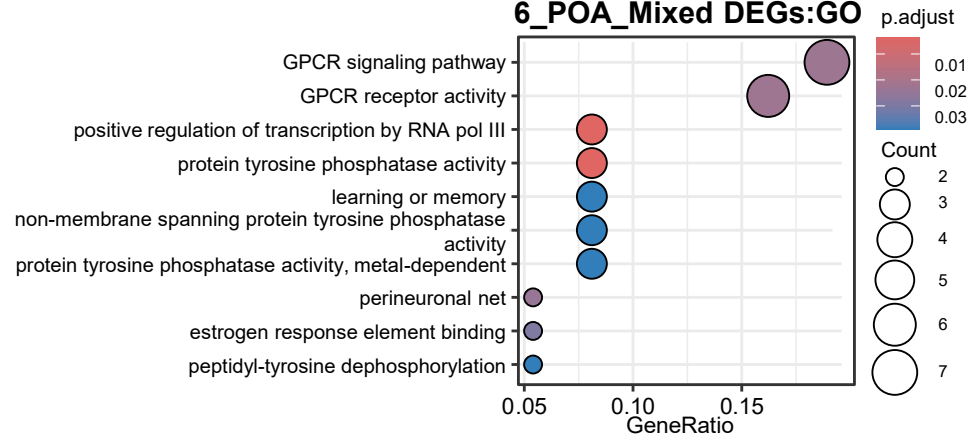


D



Supplementary Fig.5. Gene ontology (GO) analysis for the marker genes of (A) 1\_NSC, (B) 1\_gfap, and (C) 1\_shha. In A-C, color indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list. (D) Canonical and data-driven marker genes for each 1\_RGC subcluster

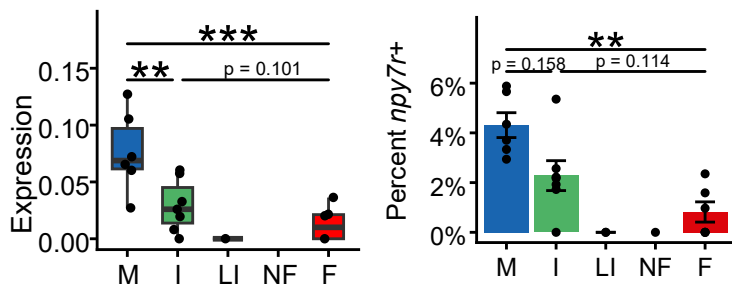
# 6\_POA\_Mixed DEGs:GO



Supplementary Fig.6. GO analysis of all of the DEGs in 6\_POA\_Mixed. Ccolor indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list.

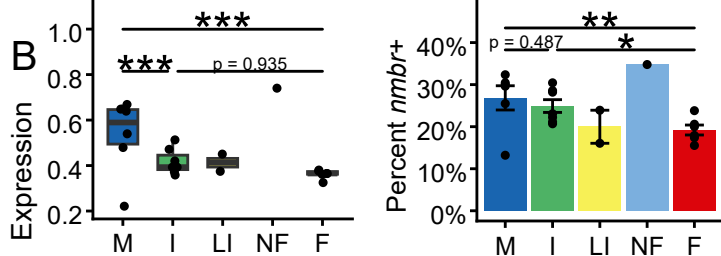
# A

## *npy7r*



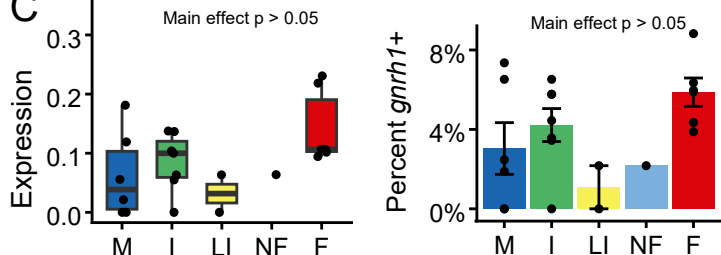
# B

## *nmb*



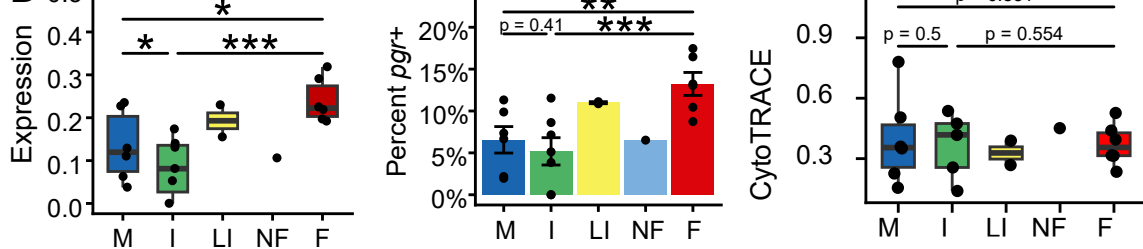
# C

## *gnrh1*



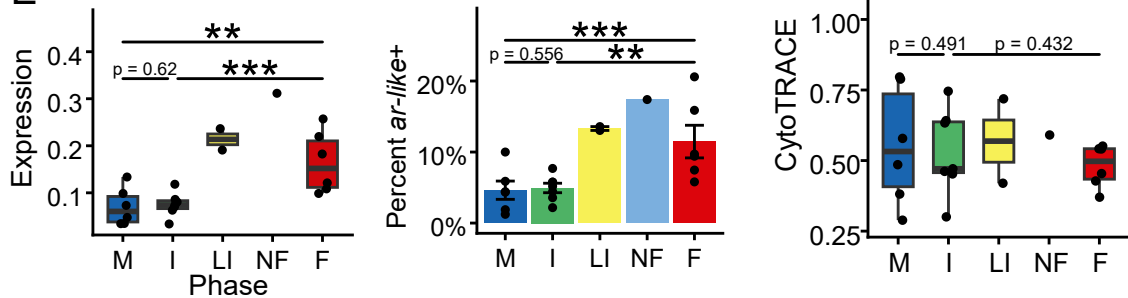
# D

## *pgr*



# E

## *ar-like*



Supplementary Fig.7. Expression (left) and percent of 6\_POA\_Mixed (right) expressing (A) *np7r*, (B) *nmbr*, (C) *gnrh1*. Expression (left) and percent of 6\_POA\_Mixed (middle), and CytoTRACE score (right) of cells expressing (D) *pgr*, (E), *ar-like*. Points indicate the mean for each individual. Asterisks: “\*”:  $p < 0.05$ , “\*\*”:  $p < 0.01$ , “\*\*\*”:  $p < 0.001$ .